3.2.2 Comparison of Electromagnet & Permanent Magnet.

ELECTRO MAGNET	PERMANENT MAGNET
1. Polarity can be changed easily.	Polarity cannot be changed easily.
2. Strength can be varied.	strength cannot be varied.
3. More cost.	Less cost
4. Suitable for motor & generator of large size.	Not suitable for large size.
5. Used in electric bells, signals, escalators, cranes.	Not used in any of these.
6. Cannot be used in navigation	Mostly used in navigation as magnetic needle
7. Cannot be used in cycle and Motor cycle dynamo.	Used in cycle and Motor cycle dynamo.

3.3 MAGNETIC MATRIALS

Magnetic materials are classified based on permeability property by three types.

a) Dia-Magnetic Materials
b) Para Magnetic Materials
c) Ferro Magnetic Materials

a) Dia – Magnetic Materials

- The materials which are repelled by a magnet are known as dia-magnetic materials. Ex: zinc, mercury, lead, sulphur, copper, silver, Bismuth, wood, etc.
- The permeability value of these materials is less than one.

b) Para Magnetic Materials

- The materials which are not strongly attracted by a magnet are known as paramagnetic materials. Ex: (aluminium, tin, platinum, magnesium etc.).
- The permeability value of these materials is just greater than one.

c) Ferro -Magnetic Materials

- The materials which are strongly attracted by a magnet are known as ferromagnetic materials. Ex: (iron, steel, nickel, cobalt, etc.)
- The permeability value of these materials is very high (varies from several hundreds to thousands).
- Materials which are easily magnetized with a high relative permeability, low coercive force (small hysteresis) are called soft ferromagnetic materials.
- Materials which are difficult to magnetize, but retain magnetism with great tenacity, with low relative permeability, high coercive force are called hard ferromagnetic materials.

3.3.1 Magnetic Terms and Properties

a) Magnetic Field:

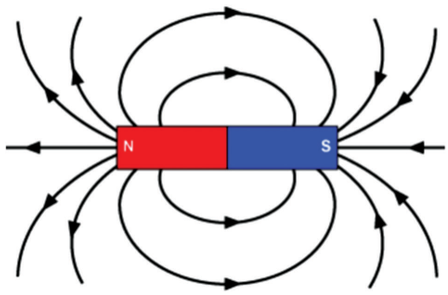


Fig 3.5 *Magnetic Field Lines*

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